

Review

Coupling Nexus and Circular Economy to Decouple Carbon Emissions from Economic Growth

Mohammed Sakib Uddin ¹, Khaled Mahmud ¹, Bijoy Mitra ¹, Al-Ekram Elahee Hridoy ²,
Syed Masiur Rahman ^{3,*}, Md Shafiullah ^{4,*}, Md. Shafiul Alam ³, Md. Ismail Hossain ⁴
and Mohammad Sujauddin ⁵

¹ Department of Geography and Environmental Studies, University of Chittagong, Chittagong 4331, Bangladesh

² Department of Geography and Environmental Studies, University of New Mexico, Albuquerque, NM 87131, USA

³ Applied Research Center for Environment & Marine Studies, Research Institute, King Fahd University of Petroleum & Minerals, Dhahran 31261, Saudi Arabia

⁴ Interdisciplinary Research Center for Renewable Energy and Power Systems, King Fahd University of Petroleum & Minerals, Dhahran 31261, Saudi Arabia

⁵ Department of Environmental Science and Management, North South University, Dhaka 1229, Bangladesh

* Correspondence: smrahman@kfupm.edu.sa (S.M.R.); shafiullah@kfupm.edu.sa (M.S.)

Abstract: Experts have been searching for ways to mitigate the impacts of climate change on resources since the early 20th century. In response, the World Economic Forum introduced the concept of a “nexus”, which involves the simultaneous, systematic collaboration of multiple individuals or sectors, such as water, energy, and food, in order to create an integrated approach to reducing resource scarcity through a multi-disciplinary framework. In contrast, a circular economy (CE) involves restructuring material flows from a linear economic system and closing the loop on resource exploitation. Both the nexus and CE have been developed to address the overexploitation of resources, but they also contribute to the Sustainable Development Goals (SDGs) and decouple carbon emissions from economic growth. This study explores the potential of combining the nexus and CE to pursue the SDGs on a global scale. Our findings reveal significant research gaps and policy implementation challenges in developing countries, as well as the potential consequences of adopting integrative scenarios. Finally, we propose a system dynamics model as a way to address the difficulties of coupling policies and to better understand the interdependencies between different parts of the economy.

Keywords: carbon emissions; circular economy; modeling; renewable energy; sustainable development; system dynamics; water–energy–food nexus; water energy footprint



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1. Introduction

With the rising global population, the world has been facing an increasing waste crisis, inadequate freshwater distribution, arable land degradation, and energy deficiency [1]. Global resource scarcity and climatic alterations also attract crucial attention as one of the most challenging tasks for policymakers. Moreover, to ensure prolific life, humans have focused on mass resource exploitation from the dawn of civilization. Since resources are limited, this attenuation has altered resource availability and degraded existing environmental circumstances [2]. Several policies and methodologies have been proposed to eradicate it for the past few years. From those, closing the loop of resource exploitation (defined as a circular economy) and using those resources in an integral framework (defined as a nexus policy) have received the most acceptance as a solution for eradicating such global threats. The thought of using waste as a resource has been introduced in different regions, even though the most popular term was the circular economy [3,4]. Nevertheless, the CE policy was introduced in the 1950s, but the nexus concept is relatively new. It allows us to

study the interrelationship between the primary demand and creates an integrated strategy that should benefit humans and the environment [5]. With the growing waste crisis and resource depletion, connecting basic individuals has become a required task regarding sustainable water, energy, and food security [6–8].

1.1. Discerning Resource Exploitation in the Age of Anthropocene

Because of humanity's widespread impact on the planet, a new geological epoch, the Anthropocene, was postulated at the turn of the twenty-first century [9]. Even at the regional magnitude, our ancestors demonstrated influence on their surroundings, such as fire-stick cultivation and mass-fauna hunts during the end of the Pleistocene. Since then, humans have technologically advanced agriculture to ensure food security. Such activity resulted in a more motionless lifestyle, the establishment of towns, and the emergence of multifaceted civilizations that eventually spanned vast swaths of land. On the other hand, regarding these human impacts, the earth's system functioned appropriately [10]. Furthermore, agriculture caused the clearing of large parts of the land surface. Only animal and human power executed this movement since energy production was still unavailable. As a result, a trend in atmospheric carbon dioxide (CO₂) concentration began to erode regarding these early agricultural activities and had a significant effect on the functioning of the earth system [11]. Even with the impact of early agriculture, the earth system was still sequentially continuous in the Holocene period [12]. At present, agriculture takes up approximately 40% of the world's land [13], and food production accounts for up to 30% of global Greenhouse Gas (GHG) releases [14] and 70% of freshwater consumption [15]. Thus, agriculture can be stated as a significant factor in global environmental change.

1.2. Rationale for Assessing the Coupling of Nexus Approach and Circular Economy

The global population expansion, particularly within urban regions, has created severe difficulties for water, energy, and food (WEF) segments, including freshwater security, food shortage, and energy deficiency [16]. Cities, for example, currently account for 7% of world freshwater withdrawal and more than 70% of global energy utilization [17] and consume more than 50% of the worldwide food supply [18,19]. Moreover, the global urban population is expected to nearly double by 2050, with the urban population in Southeast Asian cities increasing by 44 million people per year [20]. As a result, food security, sanitation, water and energy supply, and other resources were put under strain. In addition, climate change impacts, land overuse, increasing inequality, and other urban concerns have endangered our food, water, and energy security and placed pressure on future cities globally. Human activities, including unsustainable urbanization, land cover change, fossil fuel combustion, and population growth, are weakening the environment's capacity to provide water, food, and energy security and causing climate change [10].

In response to the rise in population and industrialization, Meadows et al. (1972) warned of the resource accumulation effect [21]. Furthermore, after three decades, they again remarked that economic activities are currently surpassing significant limitations, and this overshoot will become much more pronounced in the coming decades [22]. A few years after this comment, typical worldwide food costs rose steeply, making a large amount of the world's population incapable of meeting their basic nutritional demands [23]. These rising food prices can signal increasing crude resource scarceness [24]. Therefore, it is critical to identify approaches to reduce the related cross-sectoral environmental impacts for the WEF nexus [6].

Since the last decade, it has become increasingly apparent that an integrated approach to solving security challenges connected to water, energy, and food can provide co-benefits. By considering the potential impacts on specific decision-making systems, people can develop practical solutions for all sectors under consideration [25]. Therefore, adopting nexus and the CE policy will be crucial to ensure sustainability. A nexus approach enables us to evaluate the effect (trade-offs and associates) of a proposition in one sector on the other sectors with which it is linked. It is an efficient method for managing the mutuality

between resources [26]. Ensuring global water and food security, agricultural production, and energy sufficiency necessitates a well-integrated strategy due to the interdependencies between these crucial domains [27].

On the other hand, the CE provides a closed framework concerning resource exploitation. It considers the number of resources used, reshapes the whole manufacturing process, and addresses a new carbon-free methodology. Moreover, Stahel (2016) [28] stated that the CE models concern environmental and resource protection over economic development. For example, waste, by-products, and services can be turned into a resource for other products and services in multi-user systems. Therefore, the supreme goal of CE is to balance economic advancement, preserve resources, and conserve the environment [29]. Reducing waste, maintaining the highest value of products and materials, and extracting all potential by-products from resources can minimize energy consumption and CO₂ emissions. Further, with these initiatives, applying the genesis nexus framework will assist in readily operating such multifaceted programs, interlinking the water, energy, and food sectors [30]. This recognition is likely to suggest that policymaking and decision-making for sustainability could benefit from a holistic conceptual framework that reduces trade-offs and builds synergies across sectors. It therefore helps reduce costs and increase benefits for humanity and nature [31].

2. Research Approach

This article discusses the complexity-theory-based definitions of WEF nexus interactions across social, economic, political, cultural, and ecological components and their interdependencies. The review is configurative, focusing on the breadth and character of finding ideas rather than exhaustivity, and progressive, using inductive approaches to interpret particular instances to answer issues about insights and significance to build and explore theory. Our review intends to:

- Determine the significance of combining WEF in a single structure.
- Analyze previous paradigms and perspectives of nexus study in the context of developing countries.
- Investigate the results of the coupling nexus and CE policy.

In order to carry out the systematic review, peer-reviewed accessible journal articles from ScienceDirect, Springer, and the American Chemical Society were primarily applied. We reviewed works published in peer-reviewed international scientific journals in the discipline of higher learning up to July 2022, with an emphasis on the application of nexus and CE. We primarily focused on an orderly selection of literature databases from anomalous sources to observe the above objectives. Approximately 150 journal articles (including 84 review articles) were studied systematically. The preferred keywords were based on specific terms such as “WEF nexus”, “nexus in developing countries”, “nexus and circular economy”, and “effect of the linear economy”. A coupling approach has been shown to evaluate sustainability and implement environmental security. Further, a causal loop diagram (CLD) demonstrates how multiple independent resource variables in a system are causally related. The proposed CLD in Section 5.2 has been demonstrated using VENSIM software [32]. Since nexus and circular policies are multidisciplinary subjects, the criticality of this study is reviewed in the following multiple sections.

3. Background Study

3.1. WEF Nexus Thinking

Water, energy, and food security are mutually reliant in various ways, and their relationships are often referred to as the water, energy, and food (WEF) nexus [33]. However, ‘Nexus thought’ was first formulated by the “World Economic Forum” in 2011 [34] to encourage an inseparable connection between capital used to ensure necessary access to WEF security. Regarding temporal variation, WEF demand is continually increasing. In contrast, the resources required to meet it are limited and, in many cases, disappear [35]. Therefore, the “Nexus Concept” is introduced to form a securities viewpoint addressing WEF security.

Subsequent iterations of the “Nexus Framework” contained alternate elements, such as water supplies as a core element [25], land use–water–energy [36], and food as a critical component of land–water–energy linkages [24].

The primary goal of the nexus methodology is to standardize associations and provide tools for evaluating resource consumption [37]. It is a system-wide strategy that recognizes the WEF sectors’ intrinsic resource interdependencies, maximizes trade-offs and cooperation, and considers social and environmental repercussions [38,39]. It can develop integrated policy solutions to reduce trade-offs and optimize synergies across sectors, foster mutually beneficial responses that increase the possibility for collaboration across and among all sectors, and create public–private partnerships at numerous scales [33].

3.2. Significance of Nexus Study

The significance of studying the nexus has been a focus of study for decision-making exploration in the direction of sustainable development [40]. The growing population, reckless urbanization, economic prospects, and climate change have created complicated synergies in the supply–demand management of WEF resources. Therefore, all of these resources are experiencing a decline in universal storage [34]. Further, this growing demand for crude resources drives the conversation about WEF security. According to UNESCO [41], water demand is expected to rise by 55% by 2050. Furthermore, it is predicted that approximately 57% of the world’s population will face water scarcity [42], energy use will rise by 50% [43], and food demand will rise by 60% [44]. As a result, assessing the nexus is critical since the connection between these sectors makes it increasingly difficult to meet rising demands, potentially leading to additional resource scarcity [40].

WEF are all intertwined in a nexus, and activities in one area impact the others. Water and energy are required for food production; energy is needed for water extraction, treatment, and redistribution; and water is necessary for energy generation [38]. Again, water and energy demand are influenced by food choices and farming techniques [45]. Nexus thinking is a critical tool for comprehending WEF to boost resource productivity in the direction of a green economy [6]. Because agriculture accounts for 70% of global freshwater withdrawals, effective water resource management is inextricably linked to food security. Energy is crucial in the production and distribution of food and the extraction, treatment, and delivery of water [40].

Three goals are dedicated to nexus concerns among the UN’s Sustainable Development Goals (SDGs): Food security (SDG 2– End hunger, achieve food security and improved nutrition, and promote sustainable agriculture); sustainable water management (SDG 6–Ensure availability and sustainable management of water and sanitation for all); and affordable and clean energy (SDG 7– Ensure access to reasonable, consistent, sustainable, and innovative energy for all) [46,47]. WEF is inextricably linked and plays a critical role in achieving the SDGs [47]. As a result, nexus techniques can track progress toward attaining integrated SDG implementation [48]. It can further strengthen sustainability pathways by promoting higher resource use efficiency by finding positive synergies and negative trade-offs [49].

Furthermore, nexus techniques can help to improve sustainability pathways by increasing resource efficiency [49], reducing pollution [50], and reducing waste [51]. From this perspective, identifying critical interactions, conflicting needs, and hypothetical synergies in the WEF nexus can be a crucial entry point for attaining long-term adaptation [33]. A nexus approach provides a framework for resolving resource rivalry and improving resource use efficiency. The nexus approach’s aims and ethics and the focus and solutions for climate change adaptation are inextricably linked and interwoven [33].

3.3. Circular Economy

The circular economy (CE) is a systemic framework to enhance economic expansion that simultaneously profits businesses, society, and the environment [52]. It accounts for business models that extend the life cycle of a product through redesigning, refurbishing,

and recycling materials throughout the whole supply chain. Yang et al. (2018) [52] defined CE more broadly as designing holistic and curative goods that retain their worth. The CE generates a model of manufacturing and consumption that is fundamentally distinct from the society-dominated “linear economy” model. However, fundamentally, CE accounts for a series of concepts, incorporating a closed-loop approach to reproduce goods from manufacturing or component wastage. A few such significant approaches are biomimicry [53], the blue economy [54], cradle-to-cradle [55], industrial ecology [56], laws of ecology [57], the looped and operational economy [58], and reformative design [59]. Therefore, by aggregating such paradigms, the CE can be introduced as a complex whole towards emission reduction policy in a sustainable manner.

Furthermore, the CE policy relies on four functions [60]. Initially, it designs recyclable and reusable products, considering green supply chains and production methods. Then, it develops efficient business models for companies to create commercial and corporate value. Furthermore, it connects the networks of businesses and customers who manufacture and consume the core products. Finally, the policies need close monitoring to support the market. Therefore, it accounts for the systematic methodology to decouple prosperity from the demand for resources, thereby ensuring closed loops that eventually eliminate consumed goods [61]. Thus, it increases our capacity and future generations’ ability to fulfill their needs by reducing our reliance on those resources.

3.4. Significance of Circular Economy

The CE generates a more reliable, sustainable manufacturing and consumption ideal in which crude resources are possessed in production phases longer and can be reprocessed multiple times. It generates a model of producing and consuming, fundamentally distinct from the society-dominated “linear economy” model. In addition, it equalizes economic expansion and conserves resources and the environment by reducing waste [62]. Its objective is to decouple prosperity from resource reduction, i.e., how products and services can be consumed without relying on the extraction of virgin resources, thereby enhancing closed loops that will prevent the eventual disposal of consumed goods in landfills. Moreover, socially, it promotes the sharing economy, higher employment, participatory democratic decision-making, and more efficient use of existing manufacturing material through a collaborative and community user culture rather than consumer culture. Production and consumption thus have linked “contamination transfers” in each phase to the environment. In contrast, the CE apprehends using and recycling raw virgin materials and remanufactured components as necessary. Moreover, the CE makes sustainability more efficacious [63] by addressing the SDGs. For instance, using waste materials to generate H_2 is an ideal alternative since it promotes a CE in which waste recycling helps to generate energy [8].

Rapid urbanization and industrialization are creating fast occupational transitions [64]. Even if these changes tended to be a boon for humanity, they have become a curse for the environment and society. Since such economic actions are significantly founded upon the linear economy, it creates additional pressure on natural resources [65]. In contrast, a CE facilitates a sector to utilize resources sparingly and emphasize recycling [66]. Therefore, it consents to close monitoring of resources and generates a loop of reproducing needs from manufacturing wastage. Notably, it generates a framework for how products and services can be consumed yet not depend on exploiting virgin resources. Thus, CE activity develops an eco-friendly economic infrastructure using fewer resources, improving green energy and recycling, and making sustainability more effective [67]. Researchers also revealed that CE’s impact is expanded upon social aids [52,68]. For example, the CE upsurges energy proficiency, organizes productivity and object efficiency, and enhances green energy [8]. Moreover, such renewed infrastructure can often improve the environment by reducing GHG emissions by 3%, depending on the country’s gross domestic product (GDP) [69].

4. Nexus and Circular Approach

4.1. WEF Nexus

The WEF Nexus illustrates the complexity and interdependence of global resource processes to enhance adequate resource administration. To sustain population growth, it is essential to balance the objectives of dynamic resource users while ensuring the integrity of ecosystems. Therefore, the WEF nexus can be used to illustrate the complexity of combining water, energy, and food. For instance, Sanders [70] discussed future water efficiency and opportunities regarding the water–energy nexus. To begin with, water plays a crucial role in food and energy production, as well as the preservation of habitats that support agriculture and other commercial sectors necessary for attaining food safety [71]. The second essential interface is the interaction between food and water. Power is needed for agricultural production (especially irrigation) and water supply, including extraction, purification, and water circulation [38,72]. Moreover, for future research in nexus, Zhang et al. (2019) [73] developed an integrated systematic framework of individual security approaches. One of the most significant obstacles to the nexus philosophy is a lack of comprehension of the connections between WEF resources [74].

The mutuality between food, energy, and water is multifaceted. Actions in one field usually affect one or all of them, with significant fiscal, environmental, and social consequences. Indeed, the stability of one industry cannot always be accomplished without damaging another sector [24,72,75]. Moreover, ecological footprints connected to expanded water and energy usage for agricultural production increased the external costs of water and habitats, compromising world water and food systems' health and stability and highlighting the need for organized solutions [76].

4.2. Role of WEF Nexus

The early representation of the WEF nexus framework aims to counteract environmental risk factors. These risks were defined as ecosystem or biodiversity loss, climatic catastrophes, or resource deficiency [77]. A few scientific researchers and reviewers intended to solve this resource scarcity by interconnecting individuals and addressing the nexus policy [78–80]. For instance, agriculture, one of our fundamental need practices, is widely defined as a vital affair of food security. The creation of food and other primary products for sustainable living, thus contributing to food security, is the supreme function of agriculture. Therefore, to continue an equable food supply chain, agriculture consumes approximately 70% of global water withdrawal and 30% of global energy [2]. This may increase along with the increasing global population. The three axes of a highly interconnected nexus are energy, food, and water because any activity in one sector impacts the others [74]. Thus, sectoral potentials from the WEF nexus are needed to embed the root of the social and economic supply chain.

Despite being criticized for failing to address energy, food, and land issues equally [24], through the Integrated Water Resource Management (IWRM) concept, the Global Water Partnership was also an early supporter of the nexus principle [81]. The primary focus of the idea was to build up an alignment ensuring the reasonable demand for water. Nevertheless, food and freshwater require a crucial energy resource to retain parallel equivalence in the supply chain. As a result, stakeholders mutualized SDGs, Integrated Natural Resource Management, and other sustainable strategies with the WEF nexus [47,82]. This association was in response to the growing population, resource frugality, and climate change's effects on various landscapes and to reduce carbon emissions. Furthermore, the WEF nexus considers potential scenarios and connects their concepts and resolutions to resolve future complexity [44,83]. These methodologies are the result of systematic approaches that strengthen cross-sectoral perspectives and emphasize connections between subdivisions.

Interconnecting water, food, and energy may help advance positive climatic consequences by establishing green activity. Research confirmed that water, energy, and food security are all intertwined and difficult to separate. According to researchers, any strategy that focuses on one aspect of the nexus without understanding its interconnections risks

major unintended consequences [77]. Carbon emissions can be lowered by less ground-water dependency [84] and increased biofuels [85,86]. Regions with transboundary water supervision can support management since it lessens the burden of groundwater dependency [87]. Collateral energy security will avert economic risks caused by energy shortages, which will have an adverse effect on growth and social stability. Retaining carbon can prevent climate change and sustain prolific health. Furthermore, reusing materials more and more may activate sound economic stability and diminish the price hike of agricultural products. Whereas a price hike is demonstrated as an opposite precedent of sustainable life [24], the nexus approach anticipates the proper use of resources and ensures sustainable life. Again, some authors suggest that the nexus approach should not be restricted to the three-pronged concept of water, food, and energy security but should also consider land, material flows, minerals, and climate change [10].

4.3. Existing Practices of WEF Nexus in Developing Countries

Following significant research and future scenarios, it is concluded that developing countries are hypothetically more vulnerable to climate risks [88–90]. Thus, Europe’s nexus approach has been widely adopted in a few developing countries on a different scale. A glimpse of instituting this study was first seen in China at state and regional levels, indicating stakeholders’ interest and operational potential [5]. Regarding the historical evolution, nexus objectives were implemented to eradicate rising water scarcity. Several regions, including India, Pakistan [2], and Jiangsu Province in China [91], use groundwater pumping, which is widely considered an unsustainable water resource extraction method. The Murcia region in southern Spain reuses more than 90% of its water for agricultural use [92]. Environmentally vulnerable areas such as the Middle East and North African (MENA) Region witnessed a sudden resource imbalance because of their intensive energy production [93]. Freshwater shortages and reducing arable soil have created South Africa’s food shortage [94]. Instead of being an integrated process, urban design, planning, and management are still performed along sectoral lines (in “silos”), which means municipalities in Southeast Asia have been unable to capitalize on potential synergies across the energy–water–food–waste (EWWF) sectors or reap the benefits of better-integrated resource management [10]. In the development of a green economy, Denmark and nations in the Rhine River Basin have launched several measures to promote interdependencies and synergies across the nexus components while reducing trade-offs [74].

Recent research distinguishes the upward trend of the development of the WEF nexus practice. Investigating the WEF practice on a small scale, Zhang et al. (2018) [95] demonstrated a few graphs where recent trends indicate improvement in ensuring water security. To decrease groundwater dependency, FAO proposed to enhance agriculture management, improving surface irrigation and its existing technologies. It will avert the extensive use of water and energy resources. In Vietnam, several reservoirs have been created alongside the Red River Basin upstream, which play a crucial role in ensuring WEF security for the locals [96]. In Luxembourg, Clark (2003) [97] projected that greywater heat recovery might reduce household fuel usage by 63%.

While there is no comprehensive energy WEF nexus policy in the EU, there are numerous interconnections linking water and agriculture policies [98] and water and energy policies [99]. Transboundary water regulation is also a decisive practice between several countries sharing a distinctive estuary. The Amu-Darya Basin can be stated as one of these paradigms, successfully administering power and food security between Afghanistan, Tajikistan, Turkmenistan, and Uzbekistan [100]. Along with water and energy security, agriculture can make much non-food production available from waste or byproducts, for instance, a sustainable source of energy [101–103] to prolong energy production. Further, Guta et al. (2017) [104] pointed out that decentralized energy solutions are a conventional process predominantly developed in rural areas of India, China, Ethiopia, and Nepal, improving their energy security. Table 1 lists a few of the efficient and effective nexus approaches that are currently being implemented in different regions.

4.4. Circular Economy Approach

The CE concept was adorned with various concepts, despite it not having a definite implementor [105]. For example, in the 1990s, this concept was familiarized as an environmental policy in Germany, and later, in China, it was announced as a “Harmonious Society” [3]. This approach was also implemented in the UK, Japan, Korea, Denmark, and more [106]. However, all the policies mentioned above were to bring about a prodigious pause on using vast resources. The concept also enhances a comprehensive variety of subjects such as thermodynamics or an ecological economy through industrial symbiosis [107], one of the concepts of the CE. Its conviction also broadens waste management and reduces GHG emissions (for example, CO₂, NO_x, water vapor, CFC, and O₃) for sustainable development in developed nations such as the US, UK, and European Union nations [1]. Nevertheless, CE is a relatively new concept in developing countries, except for G20 nations, e.g., China. Their successful implementation of CE consists of three levels, namely, micro, meso, and macro levels [3]. Su et al. (2013) [108] appreciated the gradual CE policy in China and suggested a few stratagems regarding future CE study enhancements.

Some deficiencies in understanding the synergy in most third nations can create a CE for sustainability. Developing countries, significantly suffering from low-income scenarios, face a growing waste crisis [109]. According to the World Bank Group’s report, more than 90% of solid waste is mismanaged in these low-income countries [110]. Further, a society-based linear economy comprises a substantial amount of CO₂ emission. In 2019, the world carbon project reported an annual carbon emission of 40 Giga tons CO₂; after a decade, this amount is estimated to have doubled [111]. Regarding the estimation of burning coal, an average of 608 million hectares will be burned [112] yearly at the end of the 20th century, causing an increasing amount of 412 ppm CO₂ (Ref. [113]) in the atmosphere whereas the industrial revolution consisted of its 48%. Further, Chakravarty et al. (2009) [114] revealed that three of the top five highest global carbon emitters are developing nations.

Table 1. A summary of available WEF nexus exchanges.

Relative Case Scenario	WEF tools in Songwe River Basin of Tanzania and north of Malawi [115]	The WEF nexus in the Red River Basin in Vietnam [116]	Water Energy Agriculture nexus in Central Asia [117]	Transboundary water supervision creates three major sectors, e.g., Upstream Beneficiaries, Downstream Beneficiaries, and no Beneficiaries [118]	Integrated Natural Resource Management (INRM) [119]
Current Status of Implication	Significant change was detected in WEF sectors but the ratio was uneven with energy since the key move was to improve water and food deficiency.	Application of initiatives was gradual considering regional magnitude	Improvement trends were seen regarding coping with climatic vulnerabilities	Significant trend is observed increasing energy and economic security, bilateral conflict may cause downward trends between water–agriculture nexus	Significant changes were observed from both social and ecological perspectives, more codependency required to reach decisive progress
Adopted WEF Nexus Integration Policy	The state governance issued balanced and sustainable uses of resources, significant focus on SDG2, 6, and 7 and interaction between WEF nexus and SDGs. [80,120]	Considerable integrated approaches such as “Crop per drop” and “crop per Kwh” were developed, IWRM was introduced at regional scale [121]	The complexity of Aral Basin water management and its demographic trends was further studied by EC-IFAS and UNRCCA and a scenario thinking approach was adopted towards socio-economic benefit	Transboundary water administration between Afghanistan, Tajikistan, Turkmenistan, and Uzbekistan through Rogun Hydropower Plant [100]	Significant anomaly in the field of matter-element study, developing WEF nexus approach at regional scale [122,123]
Identified Complexity	food security and water efficiency Africa	Fresh-water supply scarcity Mediterranean	Shortage of water energy, environmental degradation Aral Sea Basin (Central Asia)	Access to equitable energy and water resource Amu-Darya Basin (Central Asia)	Downward resource trend affecting upward social economy China

Furthermore, numerous policymakers and economists supported the CE structure as having a better impact on the environment, but few researchers revealed substantial methodological gaps in implementing CE policy [69,124,125]. Significant limitations can be identified, such as economic, social, cultural, regulatory, institutional, infrastructural, policy-making authorities, etc. The linear economy has been adapted since the dawning era of industrialization for the abundance of resources [126]. Thus, it has become rather impossible to replace the policy from the ground up. Consumers often compare such resources as inexpensive regarding humanoid labor. Therefore, their root economic models are built on the predominant use of virgin materials [28]. Adapting such a policy may be compromised by both the lack of knowledge and willingness to involve the CE [127]. Further, the policymakers' detachment and non-acquaintance also limit the circular regulation. While the CE framework requires high investment, virgin materials need a small venture, and thus investors need frigid results with small funding [128].

Initiating the CE and other sustainable industrial frameworks necessitates a significant change incorporating all business entities. First, industries must restructure their manufacturing strategy and increase products' life expectancy. Here, there is a considerable barrier to adopting CE policies (for instance, reusing, repairing, remanufacturing, and refurbishing) [127,129]. The overall cost of remanufacturing or refurbishing is much higher than virgin material procurement [130–132]. While remanufacturing consists of completely replacing components with newer ones, refurbishment and repair consist of a close examination of manufactured goods and compromised parts being replaced with original, certified manufacturer parts. Furthermore, as decision-making is time-consuming, it is crucial to create efficient and straightforward methods that use statistical and machine learning algorithms [133,134]. Therefore, CE strategies require high upfront investment costs, potential technical knowledge, and a longer duration to be processed. In contrast, large- and small-scale industries prefer raw materials because production can be completed in a batch manufacturing process. As a result, several researchers declared CE strategies as less economically feasible and in need of an update before industrial policy intervention [127,130,132,135].

4.5. Role of Circular Economy towards Emissions Reduction Policy

With the essentiality of waste and GHG emissions reduction as a backdrop, the CE is increasingly considered in an applied manner. Since rapid population growth has caused the necessity of economic growth [136], GHG emissions have been significantly amplified. More than 70% of the world's net energy production depends on non-renewable sources, which causes significant tropospheric air pollutant emissions [137]. In contrast, the CE breaks the chain of using vast resources and environmental limitations [138], instating a green economy or one can be explicated as an economy with less carbon emission. For example, research on the water supply in 20 urban centers around the world showed that the mean energy consumption for water unit volume was 1.33 kWh/m³. In contrast, supply-related GHG mean emissions were 0.80 kg CO₂-eq/m³ (CO₂ equivalent compared to the emissions from various GHGs) [139]. Furthermore, the CE is organized to be sustainable and evaluates technical and biological cycles.

Nevertheless, adopting the CE approach can reduce such an index. Most such CE approaches have reduced waste, energy, and emissions. For instance, organic waste recycling [66], alleviating non-point pollutants [140], and adapting the use of biogas and biofuels [141] are some indicative measurements. In South Korea's Ulsan Eco-Industrial Park, a sewage treatment framework consisting of a low-carbon source for denitrification treats wastewater at an average of 7.92 tons/day, saving 0.7 million USD annually [142]. The industrial ecology of construction distributors in the UK has reduced landfilling waste by 65% (11,256 tons). This has resulted in annual savings of approximately 4 million USD and a net CO₂ emission reduction of 1528 tons [143]. According to Eurobarometer research (2018), 41% of small and 53% of large markets in Europe reported lower production costs by employing CE principles [60]. Moreover, Ljubljana presently generates 41% less waste per

capita than Europeans [144]. Some countries, such as the Netherlands, use unconventional natural gas produced through hydraulic fracturing as a potential bridge to a low-carbon energy system [145,146].

To reduce CO₂ emissions, the Chinese plastic recycling industry contributed 14.57 million tons of plastic recycling in 2016, amounting to a 0.1% (11.1 million tons) reduction of national GHG emissions [147]. Since environmental constancy requires a greener economy, Pan et al. (2015) [148] suggest a solution to the contradiction between greening and development through efficient resource consumption and renewable energy. A productive approach that reduces post-consumer waste pollution and mitigates carbon emissions is applying the principles of the CE in this field. Reviews of organic waste management mentioned the integration of CE and organic waste management to utilize waste to recover resources. The move to the CE needs all stakeholders, including market influencers, end-users, and policymakers, to change the conventional methods of materials used to mitigate the need for primary resource exploitation and energy consumption [149].

5. Coupling Nexus and Circular Economy

5.1. Coupling Initiatives

CE policies may help evaluate and preserve models of resources of a nation and support improving and reconstructing the environment. On the other hand, implementing a nexus policy will enhance global resource security through interconnecting individual studies. The concepts of the CE and the EFWF (Energy–Water–Food–Waste) nexus are inextricably intertwined. In contrast to an unsustainable linear economy, the CE is designed to be restorative and regenerative and can be seen as a practical solution to the emerging resource scarcity that is causing rising geopolitical tensions and supply risks, all of which contribute to volatile and insecure conditions [10]. Thus, the close linkage of these studies can be easily illustrated. The coupling nexus and the CE policy may have a crucial impact on ensuring sustainability. For instance, the current nexus policy established in European Union (EU) fails to cooperate with the environmental domain for appropriate governance problems [150]. In contrast, the EU has refurbished its economic structure by applying the CE policy. Such initiatives enhanced job opportunities and less waste creation through limited resource exploitation [151]. Despite this, present EU biofuel support policies are siloed, disregarding their impact on other sectors and long-term viability [82]. Appropriate policymaking incorporating both frameworks can ensure resource security and sustainable development through less waste creation. Consequently, a coupling approach (Figure 1) is illustrated below to facilitate organizations and future researchers to assess sustainability and implement environmental security. An end-use material flow (stream ‘a’ in Figure 1) needs to be analyzed for sustainable resource procurement in the coupled nexus and CE approach. Further, the water, energy, and food sectors need to be bound in a single framework to initiate symbiotic effects that enhance the cross-sector integration of resources [152–154]. These narratives frequently portray the Nexus as a valuable or vital approach/tool that leads to a CE as an overall appealing aim. Reusing the waste output and other discharges during the supply chain will create the circularity of procured resources and extend the life cycle of a product (streams b and c in Figure 1). Here, LCA (Life Cycle Assessment) can play a significant role in identifying the environmental impact and standards of remanufacturing assessment [112,155–157]. According to the descriptive themes, the CE and Nexus are both emerging sustainability-oriented paradigms that aim to reduce resource consumption and waste generation through inter/transdisciplinary approaches [10,158,159]. Throughout manufacturing, the cumulative input–output data will demonstrate the net carbon footprint and the ecological footprint (streams d, e, and f in Figure 1). The intersection of CE and Nexus concepts is designated as the fundamental level of relationship in this interpretation.

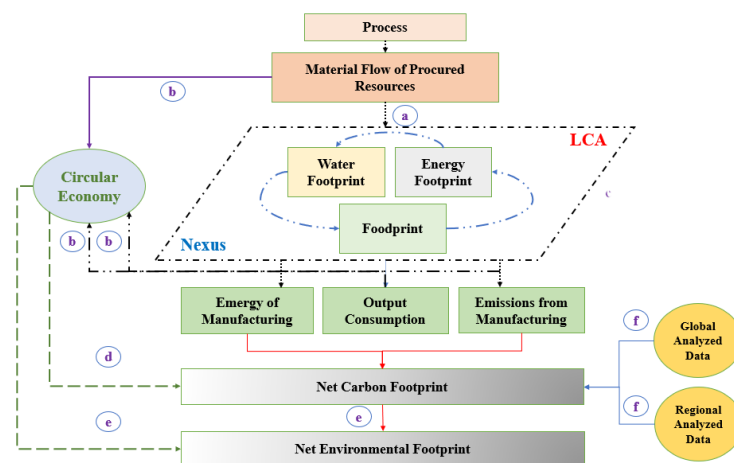


Figure 1. A methodological framework of coupling Nexus and the CE policy.

The circular policy can be incorporated into the design process with modern technologies. An equivalency evaluation is needed for integrated energy consumption throughout the restoration of the current products and the operational energy required for new production. For instance, Giama and Papadopoulos (2020) [160] demonstrated significant carbon footprint changes and socioeconomic impacts by distinguishing linear and circular frameworks. Water–energy footprints can be developed as impact-oriented parameters to support such predictions [161,162]. The water footprint involves the estimation of water across the product and service end-to-end supply chain [163,164], whereas the energy footprint evaluates the amount of land assumed to sustain CO₂ emissions [165–167]. The preceding brief description of these two notions introduces the possibility of symbiosis, where one thought supports the other, and attempts to indicate the direction of these wide liaisons [168]. Therefore, resource input and output data from a cross-sectional study must be analyzed comprehensively.

After a few decades, the present population growth rate of the world will have exhibited disastrous inflation. Supporting such a considerable population may become impossible with limited resources. WEF security will be a crucial consideration for sustaining such a large population. Therefore, policy implications are needed to ensure resource security by eradicating resource exploitation. For instance, the CE in the water sector is gaining rapid consideration because of the ongoing freshwater crisis. It deliberates a few models concerning wastewater treatment (for instance, physiochemical treatment plants, ion exchange, desalinization, and reverse osmosis [166]) to reduce surface and groundwater resource pressure. Moreover, Kakwani and Kalbar (2020) [167] revealed significant technological and methodological development worldwide by studying the CE policy in the water sector. Practicing CE initiatives through green energy production will sustainably minimize terrestrial water dependency and ensure water and crop security. It will also further act as a feature to minimize net GHG emissions from the agriculture practice. Velasco-Muñoz et al. (2021) [169] also mentioned the necessity for shifting to circular production and consumption patterns in agriculture and initiating green energy dependency [170] to reduce net GHG emissions from farming.

Furthermore, the CE law was first implemented to convert renewable energy from fossil-based energy production and enrich greener energy production [171]. Regarding the LCA as the principal tool to establish environmental sustainability, the CE policymakers introduced the waste-to-energy (WTE) policy to enhance resource sufficiency. Therefore, industrial waste, food and agricultural waste, or wastewater sludge can be efficient [8] in ensuring energy security and establishing a carbon reduction policy. Therefore, LCA is increasingly recognized as an essential and relevant tool for assessing the environmental impacts on various nexus resources [172]. Similarly, LCA is regarded as an “essential instrument” in CE for evaluating the environmental sustainability of products and technol-

ogy and the potential for increased productivity [8]. Moreover, Slorach et al. (2020) [173] distinguished a unique methodology coupling the nexus and the CE approach concerning the LCA. They characterized the impact of CE aspects on nexus policy by restoring resources from household food waste. Further, Xue et al. (2018) [16] proposed a public tool, the Urban Circular Economy Calculator (UCEC), based on the WEF nexus paradigm modeling framework. It was delivered to visualize the evaluation of multiple urban CE configurations through input from urban administrators and policymakers within the nexus. Therefore, appropriate policy enactment toward coupling assessment will increase energy, food, and environmental security.

The nexus approach is established by assessing conflicts in managing water, energy, and agricultural resources, revealing interrelationships through demand assessment, and evaluating prevailing frameworks. Therefore, close interaction is needed between the researchers and policy administrators and across complex adaptive systems from theory to practice coupling nexus and the CE. Few researchers have also proposed a few novel approaches towards coupling policies on local [174] and construction sectors [175]. However, Del Borghi et al. (2020) [6] preferred the CE policy rather than the nexus policy since the framework is more of a theoretical concept than relevant to reality. Again, removing the lack of coordination in the integrating policy can impact future resource security. Studying the barriers to the CE, Govindan and Hasanagic (2018) [176] reviewed certain barriers, including governmental and consumer perspectives. Moreover, Kaddoura and El Khatib (2017) [177] studied data gaps while integrating the nexus individuals. Below, Table 2 demonstrates the significant reviews of the present practices of nexus and circular approaches worldwide.

5.2. System Dynamics Model Corroborating Nexus Approach and Circular Economy

To initiate coupling the nexus and the CE policy, a multifaceted, enduring framework is required. Thereby, “System Dynamics Modelling” may be a significant protagonist for such an establishment. In 1960, Dr. Jay W. Forrester created a manuscript for computer modelling in which simulation and interrelation between many subjects were shown. It is argued that it is not an archetype, but rather a philosophy, a methodology showing a set of operational or hypothetical techniques or tools [183]. Thus, it can be understood as more than a modelling but rather a systematic diagram defining the correlations among many features. In comparison, coupling nexus and the CE implies adding two or more independent resource sectors in a closed chain so that the resource use is limited and reduces GHG emissions, for instance, carbon. Therefore, system dynamics can impactfully create a model where water, energy, and food interconnect with human activities and link in a chain to minimize the carbon footprint.

Figure 2 demonstrates the causal loop diagram (CLD) developed for visualizing coupled nexus and CE policy. The CLD consists of interrelated modules for modeling the interactions between demography, water, food production, and energy configurations [180]. The simulation model has been developed with VENSIM software. The core sectors consist of water, energy, and food, supported by livelihood and agricultural security, while the flows depict the impact on environmental footprint fluctuation. Water is predominantly connected to energy and food production, further ensuring livelihood and agricultural security [5,39]. Changes in livelihood regarding water demand may generate more important agricultural water requirements, contributing to an increase in cumulative water demand and possibly increasing energy demand (Figure 2) [184]. Furthermore, increased food demand will cause fluctuations in net energy production and put additional strain on resource consumption (such as water, minerals, and raw materials for manufacturing). The greater the amount of biomass produced, the more significant the level of the ecological footprint generated will be. Irrigation with treated wastewater and the creation of mass rainwater storage can eliminate such additional water demand.

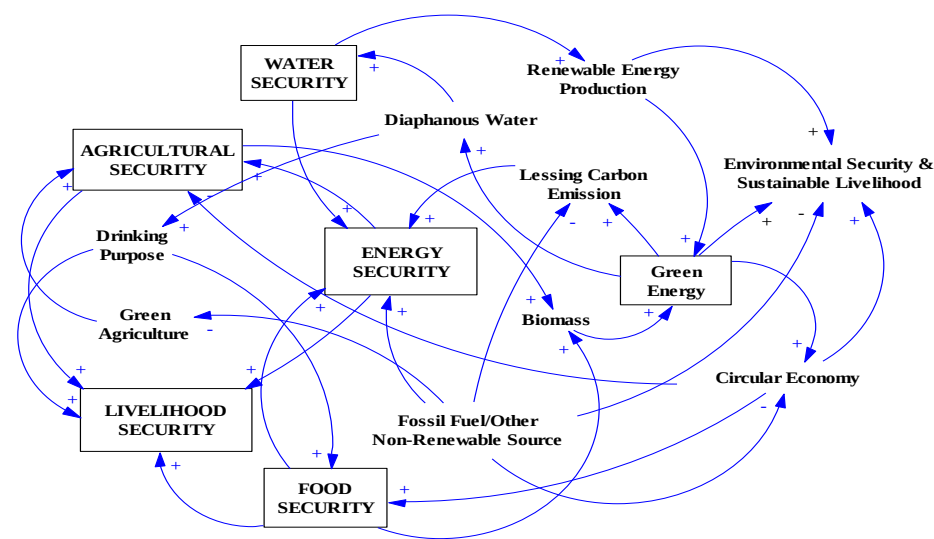


Figure 2. The conceptual causal loop diagram developed for simulating coupling of the WEF nexus and the CE policy [184].

Table 2. The nexus and circular approach.

Author	Review Topic	Key Findings
Pan et al. (2015) [148]	Strategies on implementing waste-to-energy (WTE) supply chain for the circular economy system: a review	WTE supply chain is key to maintaining energy security and controlling waste and GHG emissions. The contradiction between greening and growth needs to be resolved.
Sanders (2015) [70]	Critical review: Uncharted waters? The future of the electricity–water nexus	Integrating the water and energy production framework and changing to more sustainable energy production is needed.
Hamiche et al. (2016) [178]	A review of the water–energy nexus	From the historical perspective, proper integrative implementation is more suitable for a multidisciplinary subject such as the nexus approach. Therefore, developing the integrated methodology to enhance nexus policy regarding various dimensions is needed.
Kaddoura & El Khatib (2017) [177]	Review of water–energy–food Nexus tools to improve the Nexus modeling approach for integrated policymaking	Addressing the individual study in a single framework is needed for it to be more robust rather than consulting it as a theoretical framework.
Govindan & Hasanagic (2018) [176]	A systematic review on drivers, barriers, and practices towards a circular economy: a supply chain perspective	The government shows an effective strategy to establish CE policy; in contrast, a few perspectives acting as barriers must be resolved through the appropriate framework.
Kibler et al. (2018) [179]	Food waste and the food–energy–water nexus: A review of food waste management alternatives	Using food waste can pave the way to remedy the growing waste issue. Although, concerning environmental aspects need to draw attention, processing the waste-to-energy framework.
Albrecht et al. (2018) [5]	The Water–Energy–Food Nexus: A systematic review of methods for nexus assessment	The nexus approach has a rich theoretical framework, but poor sectoral coordination limits this approach’s advancement. Social and political aspects need the same consideration as the water–energy–food security required.
Lin et al. (2018) [180]	Improving the sustainability of organic waste management practices in the food–energy–water nexus: A comparative review of anaerobic digestion and composting	Reducing organic waste and creating energy sufficiency, several initiatives have been undertaken. Regarding LCA, anaerobic digestion is more suitable and sustainable. However, future research needs to enhance the procedure under some key factors.
Zhang et al. (2019) [73]	Food–energy–water (FEW) nexus for urban sustainability: A comprehensive review	Since nexus is a multidisciplinary concept, the implementation is hard to establish within a short period. Appropriate governance needs more effort for a secure concept and constant development.
Avtar et al. (2019) [181]	Population-urbanization-energy nexus: A review	Along with the WEF nexus, the population plays the role of concurrence of each discipline. Therefore, appropriate scientific study needs to connect the population as a resource regarding social, economic, and environmental aspects.
Paes et al. (2019) [182]	Organic solid waste management in a circular economy perspective—A systematic review and SWOT analysis	To reduce solid waste, proper concentration is needed to eradicate the limitations. Unfortunately, few methods have been proposed concerning CE and sustainability.
Sassanelli et al. (2019) [125]	Circular economy performance assessment methods: A systematic literature review	Despite CE gaining more concern daily, a few gaps are still available concerning LCA. Future researchers must provide Key Performance Indicators (KPIs) for suitable circularity.
Kakwani & Kalbar (2020) [167]	Review of Circular Economy in the urban water sector: Challenges and opportunities in India	Concerning LCA, CE in water sector implementation is an emergent policy. Therefore, developing countries need a holistic approach for its effective implementation.
Del Borghi et al. (2020) [6]	Circular economy approach to reduce water–energy–food nexus	Despite having a solid theoretical concept, poor implementation of policy hampers the possible advancement of securing resources. Therefore, CE principles are more supportive than the WEF policy.

6. Conclusions

The WEF nexus is a multidisciplinary approach developed to ensure sustainable development. Even though the essential policy was to assure appropriate development by interconnecting WEF, the chronological metamorphosis has merged the physical substances with the nexus approach. To simplify the nexus technique, researchers initially attempted to develop synergetic modules and subsequently developed a more complicated framework. On the other hand, CE is less complex to adopt in a broader sense, and thus, policymakers accept it as more practical than studying the theoretical nexus approach. Furthermore, the CE approach is relatively more acceptable than the nexus approach.

According to our review:

- Significant nexus formulations are generally regarded as less practical and frequently obstruct the fundamental path toward sustainability.
- Both natural and anthropogenic factors need to be taken into account to reduce the limitations of transferring the theoretical nexus approach to an integrated policy.
- The potential of the WEF nexus is similar to that of the CE, and coupling the nexus and CE approaches can enhance sustainability for the growing world population.
- Restructuring the manufacturing supply chain is one way to do this. It is also important to include both ideas in policymaking to ensure resource security and sustainable development.

We, therefore, presented a coupling policy incorporating sustainable resource procurement of raw materials and an end-use input assessment to allocate the environmental footprint. It will measure humanity's demands on the earth's natural systems and is crucial in comprehending climate change's pressures on those ecosystems. The framework also utilizes circular thinking by extending the life of resources and decoupling GHG emissions. Afterward, a system dynamics model was produced and constructed to demonstrate the interdependency of food, energy, and water with human activity. Lastly, it will help future academics and organizations determine how to measure sustainability and make sure the environment is safe.

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References

1. Costa, I.; Massard, G.; Agarwal, A. Waste management policies for industrial symbiosis development: Case studies in European countries. *J. Clean. Prod.* **2010**, *18*, 815–822. [CrossRef]
2. FAO. Water-Energy-Food Nexus for the Review of SDG 7. 2018. Available online: <https://sustainabledevelopment.un.org/EnergyConference/documentation> (accessed on 8 November 2021).
3. Geng, Y.; Doberstein, B. Developing the circular economy in China: Challenges and opportunities for achieving 'leapfrog development'. *Int. J. Sustain. Dev. World Ecol.* **2008**, *15*, 231–239. [CrossRef] [PubMed]

4. Halbac-Cotoara-zamfir, R.; Colantoni, A.; Mosconi, E.M.; Poponi, S.; Fortunati, S.; Salvati, L.; Gambella, F. From Historical Narratives to Circular Economy: De-Complexifying the “Desertification” Debate. *Int. J. Environ. Res. Public Health* **2020**, *17*, 5398. [CrossRef] [PubMed]
5. Albrecht, T.R.; Crootof, A.; Scott, C.A. The Water-Energy-Food Nexus: A systematic review of methods for nexus assessment. *Environ. Res. Lett.* **2018**, *13*, 043002. [CrossRef]
6. Del Borghi, A.; Moreschi, L.; Gallo, M. Circular economy approach to reduce water–energy–food nexus. *Curr. Opin. Environ. Sci. Health* **2020**, *13*, 23–28. [CrossRef]
7. Laso, J.; Margallo, M.; García-Herrero, I.; Fullana, P.; Bala, A.; Gazulla, C.; Poletini, A.; Kahhat, R.; Vázquez-Rowe, I.; Irabien, A.; et al. Combined application of Life Cycle Assessment and linear programming to evaluate food waste-to-food strategies: Seeking for answers in the nexus approach. *Waste Manag.* **2018**, *80*, 186–197. [CrossRef]
8. Sharma, S.; Basu, S.; Shetti, N.P.; Aminabhavi, T.M. Waste-to-energy nexus for circular economy and environmental protection: Recent trends in hydrogen energy. *Sci. Total Environ.* **2020**, *713*, 136633. [CrossRef]
9. Hamilton, C. The anthropocene. In *Encyclopedia of Ecology*; Elsevier: Amsterdam, The Netherlands, 2018; pp. 239–246.
10. Lehmann, S. Conceptualizing the Urban Nexus Framework for a Circular Economy: Linking Energy, Water, Food, and Waste (EWWF) in Southeast-Asian cities. In *Urban Energy Transition*; Elsevier: Amsterdam, The Netherlands, 2018; pp. 371–398.
11. Ruddiman, W.F. The anthropogenic greenhouse era began thousands of years ago. *Clim. Chang.* **2003**, *61*, 261–293. [CrossRef]
12. Steffen, W.; Persson, Å.; Deutsch, L.; Zalasiewicz, J.; Williams, M.; Richardson, K.; Crumley, C.; Crutzen, P.; Folke, C.; Gordon, L.; et al. The anthropocene: From global change to planetary stewardship. *Ambio* **2011**, *40*, 739–761. [CrossRef]
13. Foley, J.A.; DeFries, R.; Asner, G.P.; Barford, C.; Bonan, G.; Carpenter, S.R.; Chapin, F.S.; Coe, M.T.; Daily, G.C.; Gibbs, H.K.; et al. Global consequences of land use. *Science* **2005**, *309*, 570–574. [CrossRef]
14. Vermeulen, S.J.; Campbell, B.M.; Ingram, J.S.I. Climate change and food systems. *Annu. Rev. Environ. Resour.* **2012**, *37*, 195–222. [CrossRef]
15. Molden, D. *Water for Food Water for Life: A Comprehensive Assessment of Water Management in Agriculture*; International Water Management Institute: Colombo, Sri Lanka, 2007.
16. Xue, J.; Liu, G.; Casazza, M.; Ulgiati, S. Development of an urban FEW nexus online analyzer to support urban circular economy strategy planning. *Energy* **2018**, *164*, 475–495. [CrossRef]
17. FAO. Water-Energy-Food Nexus Energy Food and Agriculture Organization of the United Nations. 2019. Available online: <https://web.archive.org/web/20190311113340/http://www.fao.org/energy/water-food-energy-nexus/en/> (accessed on 12 January 2021).
18. FAO. FAOSTAT; FAO: Rome, Italy, 2015.
19. UN. *World Urbanization Prospects the 2014 Revision*; UN: New York, NY, USA, 2014.
20. GIZ. *Integrated Resource Management in Asian Cities: The Urban Nexus*; GIZ: Bonn, Germany, 2015.
21. Meadows, D.H.; Meadows, D.L.; Randers, J.; Ill, W.W.B. *Limits to Growth*; Universe Books: New York, NY, USA, 1972.
22. Meadows, D.; Randers, J.; Meadows, D. *Limits to Growth, the 30-Year Update*; Earthscan: London, UK, 2004; pp. 1–15.
23. Mohtar, R.H.; Daher, B. Water, Energy, and Food: The Ultimate Nexus. *Encycl. Agric. Food Biol. Eng.* **2012**, *1*, 1–5. [CrossRef]
24. Ringler, C.; Bhaduri, A.; Lawford, R. The nexus across water, energy, land and food (WELF): Potential for improved resource use efficiency? *Curr. Opin. Environ. Sustain.* **2013**, *5*, 617–624. [CrossRef]
25. Hoff, H. *Understanding the Nexus: Background Paper for the Bonn 2011 Nexus Conference: The Water, Energy and Food Security Nexus*; Stockholm Environment Institute: Stockholm, Sweden, 2011; p. 51. Available online: https://publications.pik-potsdam.de/pubman/faces/ViewItemFullPage.jsp?itemId=item_17278_1&view=EXPORT (accessed on 28 May 2021).
26. Kim, H.; Yoon, S.; Cho, E.; Kim, J. *A Study on Policy Directions for the Water-Food-Energy Nexus (I) Water-Energy Nexus*; Network of Institutions for Sustainable Development: Yeongi-gun, South Korea, 2015.
27. UN. *Water, Food and Energy*; UN-Water: Geneva, Switzerland, 2021.
28. Stahel, W.R. The circular economy. *Nature* **2016**, *531*, 435–438. [CrossRef]
29. Benesty, J.; Chen, J.; Huang, Y.; Cohen, I. Pearson correlation coefficient. In *Springer Topics in Signal Processing*; Springer Science and Business Media B.V.: Berlin/Heidelberg, Germany, 2009; Volume 2, pp. 1–4.
30. Reinhard, S.; Verhagen, J.; Wolters, W.; Ruben, R. *Water-Food-Energy Nexus: A Quick Scan*; Wageningen Economic Research: Wageningen, The Netherlands, 2017.
31. Bhaduri, A.; Ringler, C.; Dombrowski, I.; Mohtar, R.; Scheumann, W. Sustainability in the water–energy–food nexus. *Water Int.* **2015**, *40*, 723–732. [CrossRef]
32. Wang, L.; Zhao, Z.; Wang, X.; Xue, X. Transportation de-carbonization pathways and effect in China: A systematic analysis using STIRPAT-SD model. *J. Clean. Prod.* **2021**, *288*, 125574. [CrossRef]
33. Rasul, G.; Sharma, B. The nexus approach to water–energy–food security: An option for adaptation to climate change. *Clim. Policy* **2016**, *16*, 682–702. [CrossRef]
34. Schwab, K. *The Global Competitiveness Report 2011–2012*; World Economic Forum: Geneva, Switzerland, 2011.
35. Rockstrom, J.; Steffen, W.; Noone, K.; Persson, A.; Chapin, F.S.I.; Lambin, E.F.; Lenton, T.M.; Scheffer, M.; Folke, C.; Schellnhuber, H.J.; et al. A Safe Operating Space for Humanity: Identifying and Quantifying Planetary Boundaries that must not Be Transgressed could Help Prevent Human Activities from Causing Unacceptable Environmental Change, Argue Johan Rockstrom and

- Colleagues. *Nature* **2009**, *461*, 472–476. Available online: <https://go.gale.com/ps/i.do?p=HRCA&sw=w&issn=00280836&v=2.1&it=r&id=GALE%7CA209404108&sid=googleScholar&linkaccess=fulltext> (accessed on 23 May 2021). [CrossRef]
36. Howells, M.; Hermann, S.; Welsch, M.; Bazilian, M.; Segerström, R.; Alfstad, T.; Gielen, D.; Rogner, H.; Fischer, G.; Van Velthuizen, H.; et al. Integrated analysis of climate change, land-use, energy and water strategies. *Nat. Clim. Chang.* **2013**, *3*, 621–626. [CrossRef]
 37. Hermann, S.; Welsch, M.; Segerstrom, R.E.; Howells, M.I.; Young, C.; Alfstad, T.; Rogner, H.H.; Steduto, P. Climate, land, energy and water (CLEW) interlinkages in Burkina Faso: An analysis of agricultural intensification and bioenergy production. *Nat. Resour. Forum* **2012**, *36*, 245–262. [CrossRef]
 38. Bazilian, M.; Rogner, H.; Howells, M.; Hermann, S.; Arent, D.; Gielen, D.; Steduto, P.; Mueller, A.; Komor, P.; Tol, R.S.J.; et al. Considering the energy, water and food nexus: Towards an integrated modelling approach. *Energy Policy* **2011**, *39*, 7896–7906. [CrossRef]
 39. Mabhaudhi, T.; Nhamo, L.; Mpandeli, S.; Nhemachena, C.; Senzanje, A.; Sobratee, N.; Chivenge, P.P.; Slotow, R.; Naidoo, D.; Liphadzi, S.; et al. The water–energy–food nexus as a tool to transform rural livelihoods and well-being in Southern Africa. *Int. J. Environ. Res. Public Health* **2019**, *16*, 2970. [CrossRef]
 40. Cansino-Loeza, B.; Sánchez-Zarco, X.G.; Mora-Jacobo, E.G.; Saggiante-Mauro, F.E.; González-Bravo, R.; Mahlknecht, J.; Ponce-Ortega, J.M. Systematic Approach for Assessing the Water-Energy-Food Nexus for Sustainable Development in Regions with Resource Scarcities. *ACS Sustain. Chem. Eng.* **2020**, *8*, 13734–13748. [CrossRef]
 41. UNESCO. *IHP-VIII: Water Security: Responses to Regional and Global Challenges (2014–2021)*; UNESCO Digital Library: Paris, France, 2020; Available online: <https://unesdoc.unesco.org/ark:/48223/pf0000225103> (accessed on 23 November 2021).
 42. Boretti, A.; Rosa, L. Reassessing the projections of the World Water Development Report. *npj Clean Water* **2019**, *2*, 15. [CrossRef]
 43. Energy Information Administration. *International Energy Outlook 2021*; Energy Information Administration: Washington, DC, USA, 2021. Available online: https://www.eia.gov/outlooks/ieo/pdf/IEO2021_Narrative.pdf (accessed on 23 November 2021).
 44. FAO. *The Water-Energy-Food Nexus A New Approach in Support of Food Security and Sustainable Agriculture*; UN: Rome, Italy, 2014.
 45. Rasul, G. Managing the food, water, and energy nexus for achieving the Sustainable Development Goals in South Asia. *Environ. Dev.* **2016**, *18*, 14–25. [CrossRef]
 46. Mohtar, R. *The Importance of the Water-Energy-Food Nexus in the Implementation of the Sustainable Development Goals (SDGs)*; OCP Policy Center: Rabat, Morocco, 2016.
 47. Saladini, F.; Betti, G.; Ferragina, E.; Bouraoui, F.; Cupertino, S.; Canitano, G.; Gigliotti, M.; Autino, A.; Pulselli, F.M.; Riccaboni, A.; et al. Linking the water-energy-food nexus and sustainable development indicators for the Mediterranean region. *Ecol. Indic.* **2018**, *91*, 689–697. [CrossRef]
 48. Liu, J.; Hull, V.; Godfray, H.C.J.; Tilman, D.; Gleick, P.; Hoff, H.; Pahl-Wostl, C.; Xu, Z.; Chung, M.G.; Sun, J.; et al. Nexus approaches to global sustainable development. *Nat. Sustain.* **2018**, *1*, 466–476. [CrossRef]
 49. Biswas, A.K. Integrated Water Resources Management: Is It Working? *Int. J. Water Resour. Dev.* **2008**, *24*, 5–22. [CrossRef]
 50. Li, X.; Feng, K.; Siu, Y.L.; Hubacek, K. Energy-water nexus of wind power in China: The balancing act between CO₂ emissions and water consumption. *Energy Policy* **2012**, *45*, 440–448. [CrossRef]
 51. Amón, R.; Maulhardt, M.; Wong, T.; Kazama, D.; Simmons, C.W. Waste heat and water recovery opportunities in California tomato paste processing. *Appl. Therm. Eng.* **2015**, *78*, 525–532. [CrossRef]
 52. Geissdoerfer, M.; Savaget, P.; Bocken, N.M.P.; Hultink, E.J. The Circular Economy—A new sustainability paradigm? *J. Clean. Prod.* **2017**, *143*, 757–768. [CrossRef]
 53. Benyus, J.M. *Biomimicry: Innovation Inspired by Nature*; Harper Perennial: New York, NY, USA, 1997.
 54. Pauli, G. *The Blue Economy: 10 Years, 100 Innovations, 100 Million Jobs*; Paradigm Publication: Taos, NM, USA, 2010.
 55. McDonough, W.; Braungart, M. *Cradle to Cradle: Remaking the Way We Make Things*; North Point Press: New York, NY, USA, 2002.
 56. Graedel, T.E. On the concept of industrial ecology. *Annu. Rev. Energy Environ.* **1996**, *21*, 69–98. [CrossRef]
 57. Commoner, B. *The Closing Circle: Nature, Man, and Technology*; Random House: New York, NY, USA, 1997.
 58. Stahel, W.R. *The Performance Economy*, 2nd ed.; Springer: Berlin/Heidelberg, Germany, 2010; pp. 1–349. [CrossRef]
 59. Lyle, J. *Regenerative Design for Sustainable Development*; The Wiley Series in Sustainable Design (USA): New York, NY, USA, 1996.
 60. Geng, Y.; Sarkis, J.; Bleischwitz, R. How to globalize the circular economy. *Nature* **2019**, *565*, 153–155. [CrossRef]
 61. Morsetto, P. Targets for a circular economy. *Resour. Conserv. Recycl.* **2020**, *153*, 104553. [CrossRef]
 62. Chen, T.L.; Kim, H.; Pan, S.Y.; Tseng, P.C.; Lin, Y.P.; Chiang, P.C. Implementation of green chemistry principles in circular economy system towards sustainable development goals: Challenges and perspectives. *Sci. Total Environ.* **2020**, *716*, 136998. [CrossRef]
 63. Sauvé, S.; Bernard, S.; Sloan, P. Environmental sciences, sustainable development and circular economy: Alternative concepts for trans-disciplinary research. *Environ. Dev.* **2016**, *17*, 48–56. [CrossRef]
 64. Dribe, M.; Helgertz, J.; van de Putte, B. Did social mobility increase during the industrialization process? A micro-level study of a transforming community in southern Sweden 1828–1968. *Res. Soc. Stratif. Mobil.* **2015**, *41*, 25–39. [CrossRef]
 65. Mathews, J.A.; Tan, H. Progress Toward a Circular Economy in China. *J. Ind. Ecol.* **2011**, *15*, 435–457. [CrossRef]
 66. Lacy, P.; Rutqvist, J. *Waste to Wealth: The Circular Economy Advantage*; Palgrave Macmillan: London, UK, 2016.
 67. Omer, A.M. Energy, environment and sustainable development. *Renew. Sustain. Energy Rev.* **2008**, *12*, 2265–2300. [CrossRef]
 68. Jawahir, I.S.; Bradley, R. Technological Elements of Circular Economy and the Principles of 6R-Based Closed-loop Material Flow in Sustainable Manufacturing. In *Procedia CIRP*; Elsevier B.V.: Amsterdam, The Netherlands, 2016; Volume 40, pp. 103–108.

69. Murray, A.; Skene, K.; Haynes, K. The Circular Economy: An Interdisciplinary Exploration of the Concept and Application in a Global Context. *J. Bus. Ethics* **2017**, *140*, 369–380. [\[CrossRef\]](#)
70. Sanders, K.T. Critical review: Uncharted waters? the future of the electricity-water nexus. *Environ. Sci. Technol.* **2015**, *49*, 51–66. [\[CrossRef\]](#) [\[PubMed\]](#)
71. Rosegrant, M.W.; Cline, S.A. Global Food Security: Challenges and Policies. *Science* **2003**, *302*, 1917–1919. [\[CrossRef\]](#) [\[PubMed\]](#)
72. Stanaway, J.D.; Afshin, A.; Gakidou, E.; Lim, S.S.; Abate, D.; Abate, K.H.; Abbafati, C.; Abbasi, N.; Abbastabar, H.; Abd-Allah, F.; et al. Global, regional, and national comparative risk assessment of 84 behavioural, environmental and occupational, and metabolic risks or clusters of risks for 195 countries and territories, 1990–2017: A systematic analysis for the Global Burden of Disease Study 2017. *Lancet* **2018**, *392*, 1923–1994. [\[CrossRef\]](#)
73. Zhang, P.; Zhang, L.; Chang, Y.; Xu, M.; Hao, Y.; Liang, S.; Liu, G.; Yang, Z.; Wang, C. Food-energy-water (FEW) nexus for urban sustainability: A comprehensive review. *Resour. Conserv. Recycl.* **2019**, *142*, 215–224. [\[CrossRef\]](#)
74. Brandoni, C.; Bošnjakovic, B. Energy, food and water nexus in the European Union: Towards a circular economy. *Proc. Inst. Civ. Eng. Energy* **2018**, *171*, 140–144. [\[CrossRef\]](#)
75. Newell, P.T.; Gjerloev, J.W. Substorm and magnetosphere characteristic scales inferred from the SuperMAG auroral electrojet indices. *J. Geophys. Res. Space Phys.* **2011**, *116*, 1–15. [\[CrossRef\]](#)
76. Khan, S.; Hanjra, M.A. Footprints of water and energy inputs in food production—Global perspectives. *Food Policy* **2009**, *34*, 130–140. [\[CrossRef\]](#)
77. Bizikova, L.; Roy, D.; Swanson, D.; David, H.; Mccandless, V.M. The Water-Energy-Food Security Nexus: Towards a Practical Planning and Decision-Support Framework for Landscape Investment and Risk Management. 2013. Available online: www.iisd.org (accessed on 22 May 2021).
78. Amjath-Babu, T.S.; Sharma, B.; Brouwer, R.; Rasul, G.; Wahid, S.M.; Neupane, N.; Bhattarai, U.; Sieber, S. Integrated modelling of the impacts of hydropower projects on the water-food-energy nexus in a transboundary Himalayan river basin. *Appl. Energy* **2019**, *239*, 494–503. [\[CrossRef\]](#)
79. Hussien, W.A.; Memon, F.A.; Savic, D.A. An integrated model to evaluate water-energy-food nexus at a household scale. *Environ. Model. Softw.* **2017**, *93*, 366–380. [\[CrossRef\]](#)
80. Nhamo, L.; Mabhaudhi, T.; Mpandeli, S.; Dickens, C.; Nhemachena, C.; Senzanje, A.; Naidoo, D.; Liphadzi, S.; Modi, A.T. An integrative analytical model for the water-energy-food nexus: South Africa case study. *Environ. Sci. Policy* **2020**, *109*, 15–24. [\[CrossRef\]](#)
81. Cook, B.R.; Spray, C.J. Ecosystem services and integrated water resource management: Different paths to the same end? *J. Environ. Manag.* **2012**, *109*, 93–100. [\[CrossRef\]](#) [\[PubMed\]](#)
82. Stephan, R.M.; Mohtar, R.H.; Daher, B.; Irujo, A.E.; Hillers, A.; Ganter, J.C.; Karlberg, L.; Martin, L.; Nairizi, S.; Rodriguez, D.J.; et al. Water–energy–food nexus: A platform for implementing the Sustainable Development Goals. *Water Int.* **2018**, *43*, 472–479. [\[CrossRef\]](#)
83. Saidmamatov, O.; Rudenko, I.; Pfister, S.; Koziel, J. Water–Energy–Food Nexus Framework for Promoting Regional Integration in Central Asia. *Water* **2020**, *12*, 1896. [\[CrossRef\]](#)
84. Miller-Robbie, L.; Ramaswami, A.; Amerasinghe, P. Wastewater treatment and reuse in urban agriculture: Exploring the food, energy, water, and health nexus in Hyderabad, India. *Environ. Res. Lett.* **2017**, *12*, 075005. [\[CrossRef\]](#)
85. Woolf, D.; Lehmann, J.; Lee, D.R. Optimal bioenergy power generation for climate change mitigation with or without carbon sequestration. *Nat. Commun.* **2016**, *7*, 13160. [\[CrossRef\]](#)
86. Khan, Z.A.; Chowdhury, S.R.; Mitra, B.; Mozumder, M.S.; Elhaj, A.I.; Salami, B.A.; Rahman, M.M.; Rahman, S.M. Analysis of industrial symbiosis case studies and its potential in Saudi Arabia. *J. Clean. Prod.* **2022**, *385*, 135536. [\[CrossRef\]](#)
87. Xenarios, S.; Shenhav, R.; Abdullaev, I.; Mastellari, A. Current and future challenges of water security in central Asia. In *Water Resources Development and Management*; Springer: Berlin/Heidelberg, Germany, 2018; pp. 117–142.
88. Mertz, O.; Halsnæs, K.; Olesen, J.E.; Rasmussen, K. Adaptation to Climate Change in Developing Countries. *Environ. Manag.* **2009**, *43*, 743–752. [\[CrossRef\]](#)
89. Mirza, M.M.Q. Climate change and extreme weather events: Can developing countries adapt? *Clim. Policy* **2003**, *3*, 233–248. [\[CrossRef\]](#)
90. Thornton, P.K.; Gerber, P.J. Climate change and the growth of the livestock sector in developing countries. *Mitig. Adapt. Strat. Glob. Chang.* **2010**, *15*, 169–184. [\[CrossRef\]](#)
91. Yi, J.; Guo, J.; Ou, M.; Pueppke, S.G.; Ou, W.; Tao, Y.; Qi, J. Sustainability assessment of the water-energy-food nexus in Jiangsu Province, China. *Habitat Int.* **2020**, *95*, 102094. [\[CrossRef\]](#)
92. Navarro, T. Water reuse and desalination in Spain—Challenges and opportunities. *J. Water Reuse Desalination* **2018**, *8*, 153–168. [\[CrossRef\]](#)
93. Pachauri, R.K.; Meyer, L.A.; Core Writing Team. *Climate Change 2014 Synthesis Report*; IPCC: Geneva, Switzerland, 2014.
94. Simpson, G.B.; Jewitt, G.P.W. The development of the water-energy-food nexus as a framework for achieving resource security: A review. *Front. Environ. Sci.* **2019**, *7*, 8. [\[CrossRef\]](#)
95. Zhang, C.; Chen, X.; Li, Y.; Ding, W.; Fu, G. Water-energy-food nexus: Concepts, questions and methodologies. *J. Clean. Prod.* **2018**, *195*, 625–639. [\[CrossRef\]](#)
96. Yu, B.; Zhu, T.; Breisinger, C.; Manh Hai, N. *Examine the Agriculture, Poverty, and Climate Change Nexus in Vietnam*; AgEcon Search: St. Paul, MN, USA, 2012. [\[CrossRef\]](#)

97. Clark, W.C. From the Environment and Human Security to Sustainable Security and Development. *J. Hum. Dev.* **2003**, *4*, 289–313. [CrossRef]
98. European Court of Auditors. *Integration of EU Water Policy Objectives with the CAP: A Partial Success*; European Court of Auditors: Luxembourg, 2014.
99. EIP. *The European Innovation Partnership on Water—Water, Energy Nexus Priorities*; EIP: Brussels, Belgium, 2018.
100. Jalilov, S.-M.; Varis, O.; Keskinen, M. Sharing Benefits in Transboundary Rivers: An Experimental Case Study of Central Asian Water-Energy-Agriculture Nexus. *Water* **2015**, *7*, 4778–4805. [CrossRef]
101. Mander, Ü.; Mikk, M.; Kylvik, M. Ecological and low intensity agriculture as contributors to landscape and biological diversity. *Landsc. Urban Plan.* **1999**, *46*, 169–177. [CrossRef]
102. Nguyen, T.A.H.; Ngo, H.H.; Guo, W.S.; Zhang, J.; Liang, S.; Lee, D.J.; Nguyen, P.D.; Bui, X.T. Modification of agricultural waste/by-products for enhanced phosphate removal and recovery: Potential and obstacles. *Bioresour. Technol.* **2014**, *169*, 750–762. [CrossRef]
103. Xiao, X.; Roh, B.M.; Zhu, F. Strength Enhancement in Fused Filament Fabrication via the Isotropy Toolpath. *Appl. Sci.* **2021**, *11*, 6100. [CrossRef]
104. Guta, D.D.; Jara, J.; Adhikari, N.P.; Chen, Q.; Gaur, V.; Mirzabaev, A. Assessment of the Successes and Failures of Decentralized Energy Solutions and Implications for the Water–Energy–Food Security Nexus: Case Studies from Developing Countries. *Resources* **2017**, *6*, 24. [CrossRef]
105. Winans, K.; Kendall, A.; Deng, H. The history and current applications of the circular economy concept. *Renew. Sustain. Energy Rev.* **2017**, *68*, 825–833. [CrossRef]
106. Yuan, Z.; Bi, J.; Moriguichi, Y. The circular economy: A new development strategy in China. *J. Ind. Ecol.* **2006**, *10*, 4–8. [CrossRef]
107. Chertow, M.; Ehrenfeld, J. Self-Organizing Systems Organizing Self-Organizing Systems: Toward a Theory of Industrial Symbiosis. *J. Ind. Ecol.* **2012**, *16*, 13–27. [CrossRef]
108. Su, B.; Heshmati, A.; Geng, Y.; Yu, X. A review of the circular economy in China: Moving from rhetoric to implementation. *J. Clean. Prod.* **2013**, *42*, 215–227. [CrossRef]
109. Wilson, E. *Managing the Emerging Waste Crisis in Developing Countries' Large Cities*; IDS Policy Briefing 86; IDS: Brighton, UK, 2015; Available online: <https://opendocs.ids.ac.uk/opendocs/handle/20.500.12413/5786> (accessed on 16 April 2022).
110. Kaza, S.; Yao, L.C.; Bhada-Tata, P.; Van Woerden, F. *What a Waste 2.0: A Global Snapshot of Solid Waste Management to 2050*; World Bank: Washington, DC, USA, 2018.
111. Magazzino, C.; Mele, M.; Schneider, N.; Sarkodie, S.A. Waste generation, wealth and GHG emissions from the waste sector: Is Denmark on the path towards circular economy? *Sci. Total Environ.* **2021**, *755*, 142510. [CrossRef] [PubMed]
112. Mouillot, F.; Field, C.B. Fire history and the global carbon budget: A 1×1 fire history reconstruction for the 20th century. *Glob. Chang. Biol.* **2005**, *11*, 398–420. [CrossRef]
113. Le Quéré, C.; Andres, R.J.; Boden, T.; Conway, T.; Houghton, R.A.; House, J.I.; Marland, G.; Peters, G.P.; Van Der Werf, G.R.; Ahlström, A.; et al. The global carbon budget 1959–2011. *Earth Syst. Sci. Data* **2013**, *5*, 165–185. [CrossRef]
114. Chakravarty, S.; Chikkatur, A.; De Coninck, H.; Pacala, S.; Socolow, R.; Tavoni, M. Sharing global CO₂ emission reductions among one billion high emitters. *Proc. Natl. Acad. Sci. USA* **2009**, *106*, 11884–11888. [CrossRef]
115. Jalilov, S.M.; Amer, S.A.; Ward, F.A. Managing the water-energy-food nexus: Opportunities in Central Asia. *J. Hydrol.* **2018**, *557*, 407–425. [CrossRef]
116. Mabhaudhi, T.; Simpson, G.; Badenhorst, J.; Senzanje, A.; Jewitt, G.P.W.; Chimonyo, V.G.P.; Mpandeli, S.; Nhamo, L. Developing a Framework for the Water-Energy-Food Nexus in South Africa. *Clim. Chang. Water Resour. Afr.* **2021**, 407–431. [CrossRef]
117. Wang, Q.; Li, S.; He, G.; Li, R.; Wang, X. Evaluating sustainability of water-energy-food (WEF) nexus using an improved matter-element extension model: A case study of China. *J. Clean. Prod.* **2018**, *202*, 1097–1106. [CrossRef]
118. Muller, M. Scale and consequences—The limits of the river basin as a management unit. *Water Supply* **2019**, *19*, 618–625. [CrossRef]
119. Deng, H.M.; Wang, C.; Cai, W.J.; Liu, Y.; Zhang, L.X. Managing the water-energy-food nexus in China by adjusting critical final demands and supply chains: An input-output analysis. *Sci. Total Environ.* **2020**, *720*, 137635. [CrossRef] [PubMed]
120. Grafton, R.Q.; Garrick, D.; Manero, A.; Do, T.N. The Water Governance Reform Framework: Overview and Applications to Australia, Mexico, Tanzania, USA and Vietnam. *Water* **2019**, *11*, 137. [CrossRef]
121. Avellán, T.; Roidt, M.; Emmer, A.; Von Koerber, J.; Schneider, P.; Raber, W. Making the Water–Soil–Waste Nexus Work: Framing the Boundaries of Resource Flows. *Sustainability* **2017**, *9*, 1881. [CrossRef]
122. Keskinen, M.; Guillaume, J.H.A.; Kattelus, M.; Porkka, M.; Räsänen, T.A.; Varis, O. The water-energy-food nexus and the transboundary context: Insights from large Asian rivers. *Water* **2016**, *8*, 193. [CrossRef]
123. Markantonis, V.; Reynaud, A.; Karabulut, A.; El Hajj, R.; Altinbilek, D.; Awad, I.M.; Bruggeman, A.; Constantianos, V.; Mysiak, J.; Lamaddalena, N.; et al. Can the Implementation of the Water-Energy-Food Nexus Support Economic Growth in the Mediterranean Region? The Current Status and the Way Forward. *Front. Environ. Sci.* **2019**, *7*, 84. [CrossRef]
124. Ghisellini, P.; Cialani, C.; Ulgiati, S. A review on circular economy: The expected transition to a balanced interplay of environmental and economic systems. *J. Clean. Prod.* **2016**, *114*, 11–32. [CrossRef]
125. Sassanelli, C.; Rosa, P.; Rocca, R.; Terzi, S. Circular economy performance assessment methods: A systematic literature review. *J. Clean. Prod.* **2019**, *229*, 440–453. [CrossRef]

126. Bonviu, F. The European Economy: From a Linear to a Circular Economy. *Rom. J. Eur. Aff.* **2014**, *14*, 78. Available online: <https://heinonline.org/HOL/Page?handle=hein.journals/rojaeuf14&id=329&div=&collection=> (accessed on 9 April 2021).
127. Kirchherr, J.; Piscicelli, L.; Bour, R.; Kostense-Smit, E.; Muller, J.; Huibrechtse-Truijens, A.; Hekkert, M. Barriers to the Circular Economy: Evidence From the European Union (EU). *Ecol. Econ.* **2018**, *150*, 264–272. [\[CrossRef\]](#)
128. Rizos, V.; Behrens, A.; van der Gaast, W.; Hofman, E.; Ioannou, A.; Kafyeke, T.; Flamos, A.; Rinaldi, R.; Papadelis, S.; Hirschnitz-Garbers, M.; et al. Implementation of Circular Economy Business Models by Small and Medium-Sized Enterprises (SMEs): Barriers and Enablers. *Sustainability* **2016**, *8*, 1212. [\[CrossRef\]](#)
129. Xiao, X.; Joshi, S.; Cecil, J. Critical assessment of Shape Retrieval Tools (SRTs). *Int. J. Adv. Manuf. Technol.* **2021**, *116*, 3431–3446. [\[CrossRef\]](#)
130. de Jesus, A.; Antunes, P.; Santos, R.; Mendonça, S. Eco-innovation in the transition to a circular economy: An analytical literature review. *J. Clean. Prod.* **2018**, *172*, 2999–3018. [\[CrossRef\]](#)
131. Ranta, V.; Aarikka-Stenroos, L.; Ritala, P.; Mäkinen, S.J. Exploring institutional drivers and barriers of the circular economy: A cross-regional comparison of China, the US, and Europe. *Resour. Conserv. Recycl.* **2018**, *135*, 70–82. [\[CrossRef\]](#)
132. Shahbazi, S.; Wiktorsson, M.; Kurdve, M.; Jönsson, C.; Bjelkemyr, M. Material efficiency in manufacturing: Swedish evidence on potential, barriers and strategies. *J. Clean. Prod.* **2016**, *127*, 438–450. [\[CrossRef\]](#)
133. Ferreira, J.; Pinheiro, M.D.; Brito, J. de Refurbishment decision support tools review-Energy and life cycle as key aspects to sustainable refurbishment projects. *Energy Policy* **2013**, *62*, 1453–1460. [\[CrossRef\]](#)
134. Xiao, X.; Waddell, C.; Hamilton, C.; Xiao, H. Quality Prediction and Control in Wire Arc Additive Manufacturing via Novel Machine Learning Framework. *Micromachines* **2022**, *13*, 137. [\[CrossRef\]](#)
135. Xiao, X.; Joshi, S. Process planning for five-axis support free additive manufacturing. *Addit. Manuf.* **2020**, *36*, 101569. [\[CrossRef\]](#)
136. World Bank. *World Development Report 2017: Governance and the Law*; World Bank: Washington, DC, USA, 2017.
137. IEA. *OECD—World Energy Balances: Overview—Analysis*; IEA: Paris, France, 2018; Available online: <https://www.iea.org/reports/world-energy-balances-overview/oecd> (accessed on 16 March 2022).
138. Wang, N.; Ren, Y.; Zhu, T.; Meng, F.; Wen, Z.; Liu, G. Life cycle carbon emission modelling of coal-fired power: Chinese case. *Energy* **2018**, *162*, 841–852. [\[CrossRef\]](#)
139. Lee, M.; Keller, A.A.; Chiang, P.C.; Den, W.; Wang, H.; Hou, C.H.; Wu, J.; Wang, X.; Yan, J. Water-energy nexus for urban water systems: A comparative review on energy intensity and environmental impacts in relation to global water risks. *Appl. Energy* **2017**, *205*, 589–601. [\[CrossRef\]](#)
140. Li, Z.; Ma, Z.; van der Kuip, T.J.; Yuan, Z.; Huang, L. A review of soil heavy metal pollution from mines in China: Pollution and health risk assessment. *Sci. Total Environ.* **2014**, *468–469*, 843–853. [\[CrossRef\]](#) [\[PubMed\]](#)
141. Zhou, J.; Zhang, R.; Liu, F.; Yong, X.; Wu, X.; Zheng, T.; Jiang, M.; Jia, H. Biogas production and microbial community shift through neutral pH control during the anaerobic digestion of pig manure. *Bioresour. Technol.* **2016**, *217*, 44–49. [\[CrossRef\]](#) [\[PubMed\]](#)
142. Lei, J.; Zhang, X.; Zhou, Z.; Fan, S.; Jiang, B.; Tian, X. Experimental Study on the Treatment of 1, 4 butanediol Mixed Wastewater by Ferro-carbon Micro-electrolysis. *E3S Web Conf.* **2019**, *83*, 01015. [\[CrossRef\]](#)
143. Leigh, M.; Li, X. Industrial ecology, industrial symbiosis and supply chain environmental sustainability: A case study of a large UK distributor. *J. Clean. Prod.* **2015**, *106*, 632–643. [\[CrossRef\]](#)
144. Ilić, M.; Nikolić, M. Drivers for development of circular economy—A case study of Serbia. *Habitat Int.* **2016**, *56*, 191–200. [\[CrossRef\]](#)
145. Rasch, E.D.; Köhne, M. Hydraulic fracturing, energy transition and political engagement in the Netherlands: The energetics of citizenship. *Energy Res. Soc. Sci.* **2016**, *13*, 106–115. [\[CrossRef\]](#)
146. Xiao, X.; Roh, B.M.; Hamilton, C. Porosity management and control in powder bed fusion process through process-quality interactions. *CIRP J. Manuf. Sci. Technol.* **2022**, *38*, 120–128. [\[CrossRef\]](#)
147. Liu, Z.; Adams, M.; Cote, R.P.; Chen, Q.; Wu, R.; Wen, Z.; Liu, W.; Dong, L. How does circular economy respond to greenhouse gas emissions reduction: An analysis of Chinese plastic recycling industries. *Renew. Sustain. Energy Rev.* **2018**, *91*, 1162–1169. [\[CrossRef\]](#)
148. Pan, S.Y.; Du, M.A.; Huang, I.T.; Liu, I.H.; Chang, E.E.; Chiang, P.C. Strategies on implementation of waste-to-energy (WTE) supply chain for circular economy system: A review. *J. Clean. Prod.* **2015**, *108*, 409–421. [\[CrossRef\]](#)
149. Woolridge, A.C.; Ward, G.D.; Phillips, P.S.; Collins, M.; Gandy, S. Life cycle assessment for reuse/recycling of donated waste textiles compared to use of virgin material: An UK energy saving perspective. *Resour. Conserv. Recycl.* **2006**, *46*, 94–103. [\[CrossRef\]](#)
150. Voelker, T.; Blackstock, K.; Kovacic, Z.; Sindt, J.; Strand, R.; Waylen, K. The role of metrics in the governance of the water-energy-food nexus within the European Commission. *J. Rural Stud.* **2019**, *92*, 473–481. [\[CrossRef\]](#)
151. Kalmykova, Y.; Sadagopan, M.; Rosado, L. Circular economy—From review of theories and practices to development of implementation tools. *Resour. Conserv. Recycl.* **2018**, *135*, 190–201. [\[CrossRef\]](#)
152. Ravar, Z.; Zahraie, B.; Sharifinejad, A.; Gozini, H.; Jafari, S. System dynamics modeling for assessment of water–food–energy resources security and nexus in Gavkhuni basin in Iran. *Ecol. Indic.* **2020**, *108*, 105682. [\[CrossRef\]](#)
153. Daddi, T.; Nucci, B.; Iraldo, F. Using Life Cycle Assessment (LCA) to measure the environmental benefits of industrial symbiosis in an industrial cluster of SMEs. *J. Clean. Prod.* **2017**, *147*, 157–164. [\[CrossRef\]](#)
154. El-Houjeiri, H.; Monfort, J.C.; Bouchard, J.; Przesmitzki, S. Life Cycle Assessment of Greenhouse Gas Emissions from Marine Fuels: A Case Study of Saudi Crude Oil versus Natural Gas in Different Global Regions. *J. Ind. Ecol.* **2019**, *23*, 374–388. [\[CrossRef\]](#)
155. Harwood, S.A. In search of a (WEF) nexus approach. *Environ. Sci. Policy* **2018**, *83*, 79–85. [\[CrossRef\]](#)

156. Cheng, T.; Harrou, F.; Kadri, F.; Sun, Y.; Leiknes, T. Forecasting of wastewater treatment plant key features using deep learning-based models: A case study. *IEEE Access* **2020**, *8*, 184475–184485. [\[CrossRef\]](#)
157. Parsa, A.; Van De Wiel, M.J.; Schmutz, U. Intersection, interrelation or interdependence? The relationship between circular economy and nexus approach. *J. Clean. Prod.* **2021**, *313*, 127794. [\[CrossRef\]](#)
158. Parshall, L.; Gurney, K.; Hammer, S.A.; Mendoza, D.; Zhou, Y.; Geethakumar, S. Modeling energy consumption and CO₂ emissions at the urban scale: Methodological challenges and insights from the United States. *Energy Policy* **2010**, *38*, 4765–4782. [\[CrossRef\]](#)
159. Molina, A.J.; Llorens, P.; Garcia-Estringana, P.; de las Heras, M.M.; Cayuela, C.; Gallart, F.; Latron, J. Contributions of throughfall, forest and soil characteristics to near-surface soil water-content variability at the plot scale in a mountainous Mediterranean area. *Sci. Total Environ.* **2019**, *647*, 1421–1432. [\[CrossRef\]](#) [\[PubMed\]](#)
160. Giama, E.; Papadopoulos, A.M. Benchmarking carbon footprint and circularity in production processes: The case of stonewool and extruded polystyrene. *J. Clean. Prod.* **2020**, *257*, 120559. [\[CrossRef\]](#)
161. Hiloidhari, M.; Vijay, V.; Banerjee, R.; Baruah, D.C.; Rao, A.B. Energy-carbon-water footprint of sugarcane bioenergy: A district-level life cycle assessment in the state of Maharashtra, India. *Renew. Sustain. Energy Rev.* **2021**, *151*, 111583. [\[CrossRef\]](#)
162. Mekonnen, M.M.; Gerbens-Leenes, W. The Water Footprint of Global Food Production. *Water* **2020**, *12*, 2696. [\[CrossRef\]](#)
163. Lampridi, M.; Kateris, D.; Sørensen, C.G.; Bochtis, D. Energy Footprint of Mechanized Agricultural Operations. *Energies* **2020**, *13*, 769. [\[CrossRef\]](#)
164. Bello, M.O.; Solarin, S.A.; Yen, Y.Y. The impact of electricity consumption on CO₂ emission, carbon footprint, water footprint and ecological footprint: The role of hydropower in an emerging economy. *J. Environ. Manag.* **2018**, *219*, 218–230. [\[CrossRef\]](#)
165. Vlachos, D.; Aivazidou, E. Water Footprint in Supply Chain Management: An Introduction. *Sustainability* **2018**, *10*, 2045. [\[CrossRef\]](#)
166. Sonune, A.; Ghatge, R. Developments in wastewater treatment methods. *Desalination* **2004**, *167*, 55–63. [\[CrossRef\]](#)
167. Kakwani, N.S.; Kalbar, P.P. Review of Circular Economy in urban water sector: Challenges and opportunities in India. *J. Environ. Manag.* **2020**, *271*, 111010. [\[CrossRef\]](#)
168. Cwiklicki, M.; Wojnarowska, M. Circular Economy and Industry 4.0: One-Way or Two-way Relationships? *Eng. Econ.* **2020**, *31*, 387–397. [\[CrossRef\]](#)
169. Velasco-Muñoz, J.F.; Mendoza, J.M.F.; Aznar-Sánchez, J.A.; Gallego-Schmid, A. Circular economy implementation in the agricultural sector: Definition, strategies and indicators. *Resour. Conserv. Recycl.* **2021**, *170*, 105618. [\[CrossRef\]](#)
170. Silvestri, L.; Silvestri, C.; Forcina, A.; De Luca, C. A review of energy-based indicators for assessing sustainability and circular economy in the agri-food production. *Procedia Comput. Sci.* **2022**, *200*, 1756–1765. [\[CrossRef\]](#)
171. Le Quére, C.; Jackson, R.B.; Jones, M.W.; Smith, A.J.P.; Abernethy, S.; Andrew, R.M.; De-Gol, A.J.; Willis, D.R.; Shan, Y.; Canadell, J.G.; et al. Temporary reduction in daily global CO₂ emissions during the COVID-19 forced confinement. *Nat. Clim. Chang.* **2020**, *10*, 647–653. [\[CrossRef\]](#)
172. Ruiz-Salmón, I.; Margallo, M.; Laso, J.; Villanueva-Rey, P.; Mariño, D.; Quinteiro, P.; Dias, A.C.; Nunes, M.L.; Marques, A.; Feijoo, G.; et al. Addressing challenges and opportunities of the European seafood sector under a circular economy framework. *Curr. Opin. Environ. Sci. Health* **2020**, *13*, 101–106. [\[CrossRef\]](#)
173. Slorach, P.C.; Jeswani, H.K.; Cuéllar-Franca, R.; Azapagic, A. Environmental sustainability in the food-energy-water-health nexus: A new methodology and an application to food waste in a circular economy. *Waste Manag.* **2020**, *113*, 359–368. [\[CrossRef\]](#)
174. Kılış, Ş.; Kılış, B. Integrated circular economy and education model to address aspects of an energy-water-food nexus in a dairy facility and local contexts. *J. Clean. Prod.* **2017**, *167*, 1084–1098. [\[CrossRef\]](#)
175. Magrassi, F.; Del Borghi, A.; Gallo, M.; Strazza, C.; Robba, M. Optimal Planning of Sustainable Buildings: Integration of Life Cycle Assessment and Optimization in a Decision Support System (DSS). *Energies* **2016**, *9*, 490. [\[CrossRef\]](#)
176. Govindan, K.; Hasanagic, M. A systematic review on drivers, barriers, and practices towards circular economy: A supply chain perspective. *Int. J. Prod. Res.* **2018**, *56*, 278–311. [\[CrossRef\]](#)
177. Kaddoura, S.; El Khatib, S. Review of water-energy-food Nexus tools to improve the Nexus modelling approach for integrated policy making. *Environ. Sci. Policy* **2017**, *77*, 114–121. [\[CrossRef\]](#)
178. Hamiche, A.M.; Stambouli, A.B.; Flazi, S. A review of the water-energy nexus. *Renew. Sustain. Energy Rev.* **2016**, *65*, 319–331. [\[CrossRef\]](#)
179. Kibler, K.M.; Reinhart, D.; Hawkins, C.; Motlagh, A.M.; Wright, J. Food waste and the food-energy-water nexus: A review of food waste management alternatives. *Waste Manag.* **2018**, *74*, 52–62. [\[CrossRef\]](#) [\[PubMed\]](#)
180. Lin, L.; Xu, F.; Ge, X.; Li, Y. Improving the sustainability of organic waste management practices in the food-energy-water nexus: A comparative review of anaerobic digestion and composting. *Renew. Sustain. Energy Rev.* **2018**, *89*, 151–167. [\[CrossRef\]](#)
181. Avtar, R.; Tripathi, S.; Aggarwal, A.K.; Kumar, P. Population-urbanization-energy nexus: A review. *Resources* **2019**, *8*, 136. [\[CrossRef\]](#)
182. Paes, L.A.B.; Bezerra, B.S.; Deus, R.M.; Jugend, D.; Battistelle, R.A.G. Organic solid waste management in a circular economy perspective—A systematic review and SWOT analysis. *J. Clean. Prod.* **2019**, *239*, 118086. [\[CrossRef\]](#)
183. Pruyt, E. *What is System Dynamics? A Paradigmatic Inquiry*; System Dynamics Society: Nijmegen, The Netherlands, 2006.
184. Halbe, J.; Pahl-Wostl, C.; Lange, M.A.; Velonis, C. Governance of transitions towards sustainable development—The water-energy-food nexus in Cyprus. *Water Int.* **2015**, *40*, 877–894. [\[CrossRef\]](#)

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